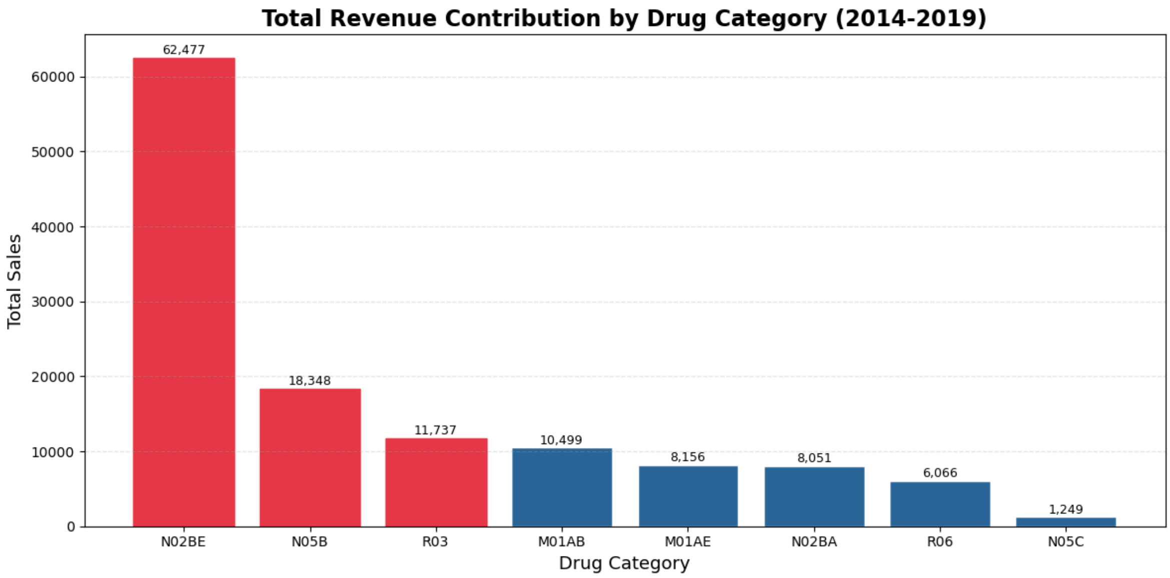
In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv(r"C:\Users\Sebastian Saldanha\OneDrive\Desktop\pharma_project\p

# Convert date and set as index
df['Datum'] = pd.to_datetime(df['Datum'], dayfirst=True)
df = df.set_index('Datum')

print("Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
```

```
Shape: (70, 8)

First 5 rows:
             M01AB    M01AE    N02BA     N02BE   N05B  N05C    R03    R06
Datum
2014-01-31  127.69   99.090  152.100   878.030  354.0    50  112.0   48.2
2014-02-28  133.32  126.050  177.000  1001.900  347.0    31  122.0   36.2
2014-03-31  137.44   92.950  147.655   779.275  232.0    20  112.0   85.4
2014-04-30  113.10   89.475  130.900   698.500  209.0    18   97.0   73.7
2014-05-31  101.79  119.933  132.100   628.780  270.0    23  107.0  123.7
```

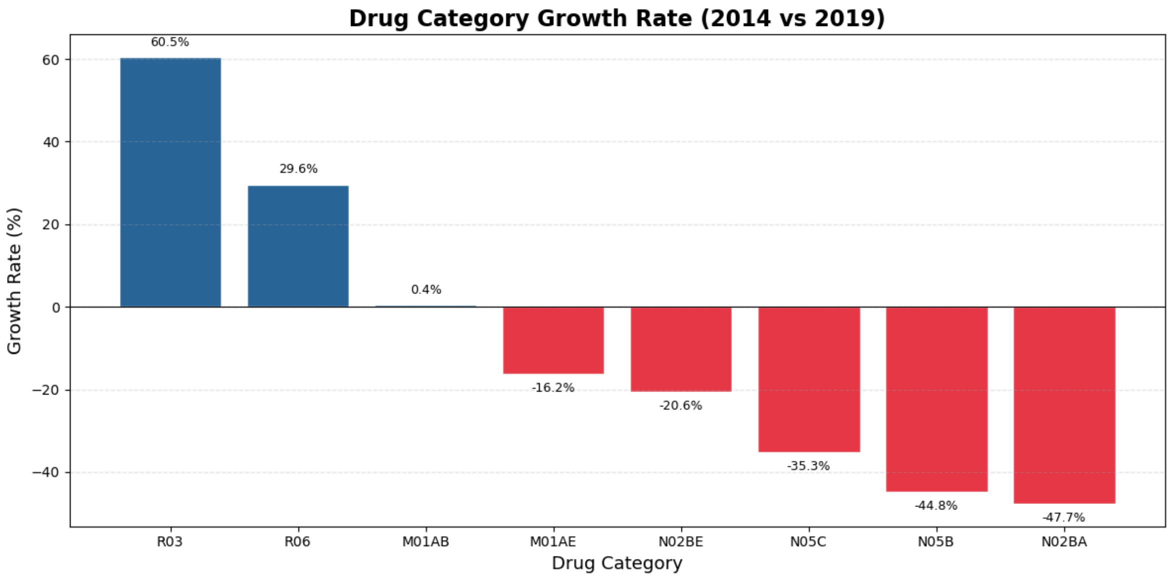
In [3]:
```
# Calculate total sales per drug category
total_sales = df.sum().sort_values(ascending=False)

plt.figure(figsize=(12, 6))
bars = plt.bar(total_sales.index, total_sales.values, color='#2A6496', edgecolor

# Highlight top 3
for i, bar in enumerate(bars):
    if i < 3:
        bar.set_color('#E63946')

plt.title('Total Revenue Contribution by Drug Category (2014-2019)', fontsize=16
plt.xlabel('Drug Category', fontsize=13)
plt.ylabel('Total Sales', fontsize=13)
plt.grid(axis='y', alpha=0.3, linestyle='--')

# Add value labels
for bar, val in zip(bars, total_sales.values):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 100,
            f'{int(val):,}', ha='center', va='bottom', fontsize=9)

plt.tight_layout()
plt.show()

print("\nRevenue Share %:")
for drug, val in total_sales.items():
    print(f"{drug}: {val/total_sales.sum()*100:.1f}%")
```




```
Revenue Share %:
N02BE: 49.4%
N05B: 14.5%
R03: 9.3%
M01AB: 8.3%
M01AE: 6.4%
N02BA: 6.4%
R06: 4.8%
N05C: 1.0%
```

In [4]:

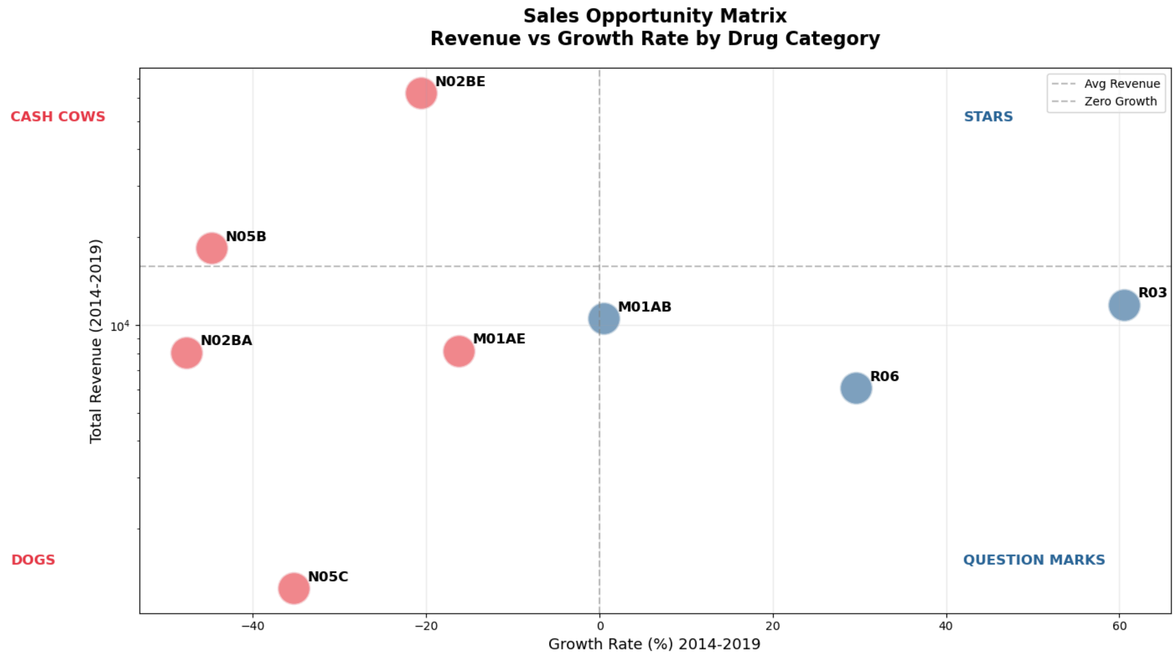
```
# Calculate yearly sales
df['Year'] = df.index.year
yearly_sales = df.groupby('Year')[['M01AB','M01AE','N02BA','N02BE','N05B','N05C'

# Calculate growth rate from 2014 to 2019
growth_rate = ((yearly_sales.loc[2019] - yearly_sales.loc[2014]) / yearly_sales.

plt.figure(figsize=(12, 6))
colors = ['#2A6496' if x > 0 else '#E63946' for x in growth_rate.values]
bars = plt.bar(growth_rate.index, growth_rate.values, color=colors, edgecolor='w

plt.title('Drug Category Growth Rate (2014 vs 2019)', fontsize=16, fontweight='b
plt.xlabel('Drug Category', fontsize=13)
plt.ylabel('Growth Rate (%)', fontsize=13)
plt.axhline(y=0, color='black', linewidth=0.8)
plt.grid(axis='y', alpha=0.3, linestyle='--')

# Add value labels
for bar, val in zip(bars, growth_rate.values):
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + (2 if val > 0 else -5),
             f'{val:.1f}%', ha='center', va='bottom', fontsize=9)

plt.tight_layout()
plt.show()

print("\nGrowth Rates:")
for drug, val in growth_rate.items():
    print(f"{drug}: {val:.1f}%")
```




```
Growth Rates:
R03: 60.5%
R06: 29.6%
M01AB: 0.4%
M01AE: -16.2%
N02BE: -20.6%
N05C: -35.3%
N05B: -44.8%
N02BA: -47.7%
```

In [8]:
```
# Bubble chart - Revenue vs Growth Rate
total_sales = df.drop(columns=['Year']).sum()
growth_rate_dict = growth_rate.to_dict()

fig, ax = plt.subplots(figsize=(14, 8))

for drug in total_sales.index:
    x = growth_rate_dict[drug]
    y = total_sales[drug]
    size = 800  # fixed bubble size for clarity
    color = '#2A6496' if x > 0 else '#E63946'
    ax.scatter(x, y, s=size, color=color, alpha=0.6, edgecolors='white', linewid
    ax.annotate(drug, (x, y), textcoords="offset points",
                xytext=(12, 6), fontsize=12, fontweight='bold')

ax.axhline(y=total_sales.mean(), color='gray', linestyle='--', alpha=0.5, label=
ax.axvline(x=0, color='gray', linestyle='--', alpha=0.5, label='Zero Growth')

ax.set_title('Sales Opportunity Matrix\nRevenue vs Growth Rate by Drug Category'
             fontsize=16, fontweight='bold', pad=20)
ax.set_xlabel('Growth Rate (%) 2014-2019', fontsize=13)
ax.set_ylabel('Total Revenue (2014-2019)', fontsize=13)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.2)
ax.set_yscale('log')  # log scale so all drugs visible

# Quadrant labels
ax.text(42, 50000, 'STARS', fontsize=12, color='#2A6496', fontweight='bold')
ax.text(-68, 50000, 'CASH COWS', fontsize=12, color='#E63946', fontweight='bold'
ax.text(42, 1500, 'QUESTION MARKS', fontsize=12, color='#2A6496', fontweight='bc
ax.text(-68, 1500, 'DOGS', fontsize=12, color='#E63946', fontweight='bold')
```

```python
plt.tight_layout()
plt.show()
```



In [9]:
```python
print("=" * 55)
print("   SALES OPTIMIZATION ANALYSIS - BUSINESS SUMMARY")
print("=" * 55)

print("""
OBJECTIVE:
Identify revenue growth opportunities and optimize
sales resource allocation across drug categories.

KEY FINDINGS:
1. N02BE (Paracetamol) = 49.4% of total revenue
   but declining at -20.6% - over-reliance risk

2. R03 (Respiratory) growing at +60.5% - highest
   growth potential, currently underinvested

3. R06 (Antihistamines) growing at +29.6% -
   second fastest growing category

4. N02BA & N05B declining 44-48% - consider
   deprioritizing sales resources here

RECOMMENDATIONS:
- Reallocate 20-30% of sales effort from declining
  drugs (N02BA, N05B) to high growth R03 and R06
- Develop R03 & R06 as next revenue pillars to
  reduce over-dependence on Paracetamol
- Potential revenue upside of 15-20% if growth
  categories are properly resourced
""")
print("=" * 55)
```

```
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   SALES OPTIMIZATION ANALYSIS - BUSINESS SUMMARY
=======================================================

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=======================================================
```

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